Yes. I'll treat the uploaded PDF as the **source for the interview notes**, and apply an **80/20 interview filter**: focus on the concepts most likely to give you the largest practical/interview payoff, while keeping the mathematical intuition, formulas, diagrams, and implementation.

The uploaded material is centered on **Expected Value, Variance, Covariance/Correlation, Law of Large Numbers, Information Theory/Entropy, and Maximum Likelihood Estimation**.

Interview Notes — Probability & Statistics for ML

0. 80/20 Priority Map

Priority	Concept	Interview importance	What you should be able to do
🔴 P0	Expected Value	Very High	Define, calculate, explain intuition
🔴 P0	Variance & Std. Dev.	Very High	Explain spread and calculate
🔴 P0	Covariance	Very High	Explain relationship between variables
🔴 P0	Correlation	Very High	Explain scale-independent relationship
🔴 P0	Independence vs correlation	Very High	Explain the common trap
🔴 P0	Law of Large Numbers	Very High	Explain why sample averages stabilize
🔴 P0	MLE	Very High	Explain likelihood → log-likelihood → optimization
🟠 P1	Linearity of expectation	High	Solve expectation problems quickly
🟠 P1	LOTUS	High	Calculate E[g(X)]
🟠 P1	Markov/Chebyshev	Medium-High	Explain concentration/bounds
🟠 P1	Entropy	High for ML	Explain uncertainty/information
🟡 P2	Continuous-variable derivations	Medium	Understand, not memorize
🟡 P2	Detailed proofs	Medium	Know intuition; derive key ones
🟡 P2	Differential entropy subtleties	Lower	Understand conceptually

---

1. Expected Value

Definition

The **expected value** of a random variable is its long-run average value.

For a discrete random variable:

$$E[X] = \sum_{x} x\, P(X = x)$$

The PDF explicitly describes it as the average value of a random variable in the long run.

Intuition

Think:

Possible outcomes

|

├— x_1 — probability p_1

├— x_2 — probability p_2

├— x_3 — probability p_3

|

▼

Weighted average

|

▼

E[X]

It is **not necessarily an outcome that can actually occur**.

Example

Suppose:

X = 10 with probability 0.2

X = 20 with probability 0.5

X = 30 with probability 0.3

Then:

$$E[X] = 10(0.2) + 20(0.5) + 30(0.3)$$
$$= 2 + 10 + 9 = 21$$

So:

Expected value = 21

Interview Example: Gambling

The PDF uses a game where:

- Heads → +$1
- Tails → −$2
- each has probability 1/2

Therefore:

$$E[X] = 1(0.5) + (-2)(0.5) = -0.5$$

So over many plays, the average winnings approach −$0.50 per round.

Interview answer

"Expected value is the probability-weighted average of a random variable's possible outcomes. It represents the long-run average, not necessarily a value that must occur in a single trial."

---

2. Expected Value of a Function — LOTUS

A very useful result is:

$$E[g(X)] = \sum_x g\,(x)P(X = x)$$

for discrete X.

For continuous X:

$$E[g(X)] = \int g\,(x)f_X(x)\,dx$$

The source calls this the **Law of the Unconscious Statistician (LOTUS)**.

Why is it useful?

Suppose:

$$X = \text{prediction error}$$

and we want:

$$E[X^2]$$

We don't need to derive the distribution of X^2.

We directly calculate:

$$E[X^2]$$

This is extremely relevant to ML because:

$$MSE = E\left[\left(\hat{Y} - Y\right)^2\right]$$

The source explicitly connects MSE to expected value.

Interview question

Q: Why does MSE use an expectation?

Answer:

Because we want the average squared prediction error over the data-generating distribution. The empirical MSE estimates the population quantity $E[(f(X) - Y)^2]$.

---

3. Linearity of Expectation 🔴

This is one of the most useful properties to memorize.

$$E[aX + bY] = aE[X] + bE[Y]$$

Importantly:

X and Y do NOT need to be independent.

The source identifies linearity as an essential property.

Example

Suppose:

$$E[X] = 10$$
$$E[Y] = 20$$

Find:

$$E[3X + 2Y + 5]$$

Use linearity:

$$E[3X + 2Y + 5] = 3E[X] + 2E[Y] + 5$$
$$= 30 + 40 + 5$$
$$= 75$$

Interview trap

Q: Do X and Y need to be independent for linearity of expectation?

No.

That's a very common interview question.

---

4. Expected Value of a Product

Here there **is** an important condition.

If X and Y are independent:

$$E[XY] = E[X]E[Y]$$

The source explicitly gives this result for independent random variables.

But in general:

$$E[XY] \neq E[X]E[Y]$$

Remember

SUM:

E[X + Y] = E[X] + E[Y]

Always works.

PRODUCT:

E[XY] = E[X]E[Y]

Needs independence.

---

5. Variance 🔴

Expected value tells us the **center**.

Variance tells us the **spread around that center**.

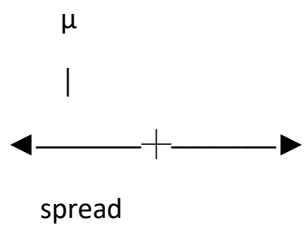
The source defines:

$$Var(X) = E[(X - \mu)^2]$$

where:

$$\mu = E[X]$$

Diagram



X values:

• • • • • •

Why square the difference?

Because without squaring:

$$E[X - \mu] = 0$$

Positive and negative deviations cancel.

Squaring prevents cancellation.

---

6. Shortcut Formula for Variance 🔴

Instead of:

$$Var(X) = E[(X - \mu)^2]$$

we can use:

$$\boxed{Var(X) = E[X^2] - E[X]^2}$$

The source derives exactly this identity.

Example

Suppose:

X = 1, 2, 3

with equal probability.

Step 1: Mean

$$E[X] = \frac{1 + 2 + 3}{3} = 2$$

Step 2: $E[X^2]$

$$E[X^2] = \frac{1^2 + 2^2 + 3^2}{3} = \frac{14}{3}$$

Step 3

$$Var(X) = \frac{14}{3} - 4 = \frac{2}{3}$$

---

7. Standard Deviation

$$Std(X) = \sqrt{Var(X)}$$

The source defines standard deviation this way.

Why do we need it?

Because variance is measured in **squared units**.

For example:

Height → meters

Variance → meters²

Standard deviation → meters

So standard deviation is easier to interpret in the original unit.

---

8. Variance of a Scaled Variable

Very important:

$$Var(aX) = a^2 Var(X)$$

The source explicitly states this property.

Example

If:

$$Var(X) = 4$$

then:

$$Var(3X) = 9(4) = 36$$

Interview trap

Some candidates answer:

$$Var(3X) = 3Var(X)$$

❌ Wrong.

It is:

$$\boxed{9Var(X)}$$

---

9. Covariance 🔴

Variance examines **one variable**.

Covariance examines **two variables**.

The source defines:

$$Cov(X,Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

Intuition



Both increase together

↓

positive covariance

If X tends to increase when Y increases:

$$Cov(X,Y) > 0$$

If X tends to increase when Y decreases:

$$Cov(X,Y) < 0$$

If there is no linear tendency:

$$Cov(X,Y) \approx 0$$

---

10. Covariance Shortcut

Memorize:

$$\boxed{Cov(X,Y) = E[XY] - E[X]E[Y]}$$

The source provides this simplified form.

Example

Suppose:

$$E[X] = 2$$
$$E[Y] = 5$$
$$E[XY] = 12$$

Then:

$$Cov(X, Y) = 12 - (2)(5)$$
$$= 2$$

Positive covariance.

---

11. Correlation 🔴

Covariance has a problem:

its magnitude depends on the units of X and Y.

Correlation normalizes covariance:

$$\boxed{Corr(X, Y) = \frac{Cov(X, Y)}{\sigma_X \sigma_Y}}$$

This is the definition given in the source.

Correlation lies between:

$$-1 \leq Corr(X, Y) \leq 1$$

Interpretation

-1 0 +1

|----------|----------|

strong negative no linear strong positive

relationship

---

12. Covariance vs Correlation

	Covariance	Correlation
Measures	Joint variation	Normalized linear relationship
Scale	Depends on units	Unitless
Range	Unbounded	−1 to +1
Formula	$E[XY] - E[X]E[Y]$	$Cov/(\sigma_X \sigma_Y)$
Interpretation	Harder	Easier

Interview answer

"Covariance tells us whether two variables move together, but its magnitude depends on their scales. Correlation normalizes covariance by the standard deviations, making it unitless and bounded between −1 and 1."

---

13. Independence vs Correlation 🔴🔥

This is an **extremely important interview trap**.

The source states:

$$X \perp Y \Rightarrow Cov(X, Y) = 0$$

and therefore:

$$Corr(X, Y) = 0$$

But:

$$Cov(X, Y) = 0 \Rightarrow / X \perp Y$$

The source explicitly emphasizes this distinction.

Classic example

Let:

$$X \in \{-, 1, 01\}$$

with each probability $1/3$.

Define:

$$Y = X^2$$

Then Y is completely determined by X.

Therefore X and Y are **not independent**.

Yet:

$$Cov(X, Y) = 0$$

The source works through exactly this example.

Memorize this:

Independence

↓

Zero covariance

BUT

Zero covariance

✗

does NOT imply

independence

---

14. Law of Large Numbers 🔴

The basic idea:

As the number of independent observations increases, the sample average approaches the population expected value.

Let:

$$\bar{X}_n = \frac{X_1 + \cdots + X_n}{n}$$

Then:

$$\bar{X}_n \to E[X]$$

in the appropriate sense.

The source illustrates this with repeated coin tosses and dice rolls.

Diagram

Number of samples

|

▼

10 → average fluctuates

100 → more stable

1000 → even more stable

10000 → approaches E[X]

Example: Dice

For a fair six-sided die:

$$E[X] = \frac{1+2+3+4+5+6}{6} = 3.5$$

If we roll the die repeatedly:

10 rolls → maybe 4.1

100 rolls → maybe 3.7

1,000 rolls → maybe 3.52

10,000 rolls → closer to 3.5

The source uses this exact intuition with dice.

---

15. Weak vs Strong Law of Large Numbers

Weak Law

The sample average converges to the mean **in probability**:

$$P(|\ \bar{X}_n - \mu\ | \geq \epsilon) \to 0$$

The source gives this formulation and proof using Chebyshev's inequality.

Strong Law

The sample average converges to the mean **almost surely**:

$$P\left(\lim_{n\to\infty} \bar{X}_n = \mu\right) = 1$$

Interview-level distinction

Weak LLN

↓

Probability of being far away → 0

Strong LLN

↓

Actual sequence converges

almost surely

You don't normally need the full proof for an ML interview unless the interviewer is testing probability theory deeply.

---

16. Markov's Inequality 🟠

For a non-negative random variable:

$$\boxed{P(X \geq t) \leq \frac{E[X]}{t}}$$

The source defines Markov's inequality this way.

Intuition

If the average is small, the probability of seeing a very large value cannot be too large.

Example:

$$E[X] = 10$$

What can we say about:

$$P(X \geq 100)?$$

Markov:

$$P(X \geq 100) \leq \frac{10}{100} = 0.1$$

So probability is at most 10%.

---

17. Chebyshev's Inequality 🟠

Chebyshev gives a bound based on variance:

$$\boxed{P(|\ X - \mu\ | \geq t) \leq \frac{\sigma^2}{t^2}}$$

The source derives it from Markov's inequality.

Intuition

Variance ↑

↓

More spread

↓

Less certainty near mean

Variance ↓

↓

Less spread

↓

More concentration near mean

Example

Suppose:

$$\mu = 100$$

$$\sigma^2 = 25$$

What is the maximum probability of being at least 10 away?

$$P(|\ X - 100\ | \geq 10) \leq \frac{25}{100} = 0.25$$

So at most 25%.

---

18. Why LLN Works

This is a nice interview-level derivation.

For:

$$\bar{X}_n = \frac{X_1 + \cdots + X_n}{n}$$

if the variables are independent:

$$Var(\bar{X}_n) = \frac{\sigma^2}{n}$$

The source derives exactly this.

Therefore:

n increases

↓

variance of average decreases

↓

sample average becomes more concentrated

↓

sample average approaches μ

This is one of the best ways to **explain LLN instead of memorizing it**.

---

19. Information Theory — Entropy 🔴

The later part of the chapter introduces information theory and entropy. The source discusses discrete entropy and then differential entropy.

For a discrete random variable:

$$\boxed{H(X) = -\sum_{x} P\,(X = x) \log P\,(X = x)}$$

Intuition

Entropy measures **uncertainty**.

Fair coin

Coin

/ \

Heads Tails

50% 50%

High uncertainty

Almost deterministic coin

Coin

/ \

Heads Tails

99% 1%

Low uncertainty

Therefore:

More uncertainty

↓

Higher entropy

More predictable

↓

Lower entropy

---

20. Why Entropy Matters in ML

Entropy is important for:

- classification
- decision trees
- information gain
- probabilistic models
- cross-entropy
- neural-network classification losses
- information theory

Key idea

If a model says:

Cat = 99%

Dog = 1%

there is relatively little uncertainty.

If:

Cat = 50%

Dog = 50%

there is more uncertainty.

---

21. Differential Entropy

For a continuous random variable:

$$\boxed{h(X) = -\int f\,(x) \log f\,(x)\, dx}$$

The source derives the continuous analogue and shows, for example, that:

$$h\big(Uniform(a, b)\big) = \log(b - a)$$

Important interview distinction

Don't say:

"Continuous entropy is always positive."

That's false.

Differential entropy can be negative.

The source explicitly notes that it can be negative.

---

22. Maximum Likelihood Estimation — MLE 🔴🔥

This is probably the **most important ML concept in the latter part of the chapter**.

Definition

MLE chooses the parameter that makes the observed data most likely.

$$\boxed{\hat{\theta} = \arg\max_{\theta} LLH(\theta; x_1; \dots; x_n)}$$

The source defines MLE this way.

---

23. MLE Intuition

Suppose you observe:

H H T H H H T H

You want to estimate:

$$p = P(Heads)$$

Possible candidates:

p = 0.2

p = 0.5

p = 0.75

p = 0.9

Which one makes the observed sequence most likely?

MLE chooses the parameter that maximizes that likelihood.

Observed data

|

▼

Choose probabilistic model

|

▼

Compute likelihood

|

▼

Maximize likelihood

|

▼

Estimated parameters $\hat{\theta}$

---

24. Coin-Toss MLE

Suppose:

Heads = H

Tails = T

and we observe:

$$H, H, H, T, T$$

There are 3 heads and 2 tails.

Likelihood:

$$L(p) = p^3(1-p)^2$$

We want:

$$\hat{p} = \arg\max_p p^3(1-p)^2$$

Taking log:

$$\log L\,(p) = 3\log p + 2\log(1-p)$$

Differentiate:

$$\frac{3}{p} - \frac{2}{1-p} = 0$$

Solving:

$$3(1-p) = 2p$$
$$3 = 5p$$
$$\boxed{\hat{p} = 0.6}$$

Which is simply:

$$\frac{\text{number of heads}}{\text{number of tosses}} = \frac{3}{5} = 0.6$$

---

25. Why Log-Likelihood?

This is a **must-know interview question**.

Suppose:

$$L(\theta) = \prod_{i=1}^{n} P_\theta\,(x_i)$$

Products can become numerically tiny.

Taking logarithms:

$$\log L\,(\theta) = \sum_{i=1}^{n} \log P_\theta\,(x_i)$$

So:

Likelihood

$p_1 \times p_2 \times p_3 \times \ldots \times p_n$

↓ log

$\log p_1 + \log p_2 + \log p_3 + ... + \log p_n$

The source explicitly notes that logarithms turn the product into a sum.

Why is maximizing log-likelihood equivalent?

Because logarithm is monotonically increasing:

$$\arg\max L(\theta) = \arg\max \log L(\theta)$$

---

26. MLE Implementation — Python

This is worth implementing because MLE is an optimization concept.

```
import numpy as np


data = np.array([1, 1, 1, 0, 0])


def log_likelihood(p, data):
    if p <= 0 or p >= 1:
        return -np.inf

    return np.sum(
        data * np.log(p)
        + (1 - data) * np.log(1 - p)
    )


candidates = np.linspace(0.01, 0.99, 1000)


best_p = max(
    candidates,
    key=lambda p: log_likelihood(p, data)
```

)

print(best_p)

You'll get a value close to:

0.6

This gives you the full engineering flow:

Data

↓

Probability model

↓

Likelihood

↓

Log likelihood

↓

Optimization

↓

Parameter estimate

---

27. MLE for Gaussian Distribution

Suppose:

$$X_1, \ldots, X_n \sim N(\mu, \sigma^2)$$

MLE produces:

$$\boxed{\hat{\mu} = \frac{1}{n}\sum_i x_i}$$

and:

$$\boxed{\hat{\sigma}^2 = \frac{1}{n}\sum_i (x_i - \hat{\mu})^2}$$

The source explicitly derives these estimates.

Important interview point

Notice the variance denominator:

$$n$$

not:

$$n-1$$

For the **MLE estimate of Gaussian variance**, the denominator is n.

This is different from the commonly used unbiased sample variance estimator.

---

28. MLE General Algorithm

Memorize this workflow:

1. Choose probability distribution

↓

2. Write likelihood L(θ)

↓

3. Take log

↓

4. Obtain log-likelihood

↓

5. Differentiate / optimize

↓

6. Find $\hat{\theta}$

↓

7. Verify it is a maximum

The source describes essentially this classical procedure.

---

29. MLE vs Probability

This distinction is commonly tested.

Probability

Parameters are fixed.

Data is variable.

θ fixed

↓

What is P(data | θ)?

Likelihood

Data is fixed.

Parameters are variable.

data fixed

↓

Which θ makes the data most likely?

So:

$$P(X \mid \theta)$$

can be viewed as a probability distribution over X when θ is fixed.

But when we observe x and vary θ:

$$L(\theta \mid x)$$

is a **likelihood function**.

---

30. MLE vs Loss Function

Very important ML connection.

MLE:

$$\max_{\theta} \log P_{\theta}\,(data)$$

Equivalent to:

$$\min_{\theta} -\log P_{\theta}\,(data)$$

So **negative log-likelihood becomes a loss function**.

Maximum Likelihood

|

▼

maximize log-likelihood

|

▼

equivalent

|

▼

minimize negative log-likelihood

|

▼

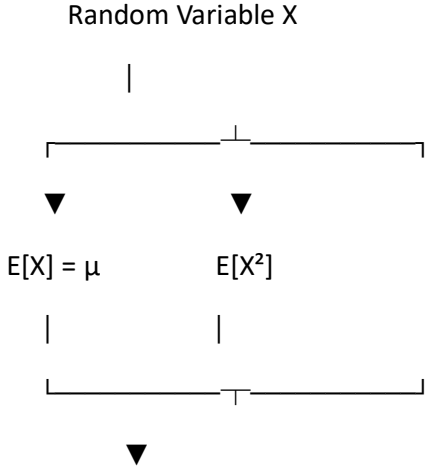
ML loss

This is the conceptual bridge between classical statistics and machine learning.

---

31. Expected Value → Variance → Covariance → MLE

You should understand the relationships rather than memorize isolated formulas.



$$\text{Var}(X) = E[X^2] - \mu^2$$

|

▼

$$\text{Std}(X) = \sqrt{\text{Var}}$$

X + Y

|

▼

Cov(X,Y)

|

▼

Normalize by σX σY

|

▼

Correlation

Then:

Probability model

|

▼

Likelihood

|

▼

Log-likelihood

|

▼

Optimization

|

▼

MLE parameters

---

32. ML Connection: MSE

The source explicitly connects MSE with expectation:

$$MSE = E[(f(X) - Y)^2]$$

So when an interviewer asks:

"Why is MSE related to probability?"

Answer:

"MSE is the empirical estimate of the expected squared prediction error. In probabilistic terms, the population objective is $E[(f(X) - Y)^2]$."

---

33. ML Connection: Model Evaluation

Suppose you have prediction errors:

-2, -1, 0, 1, 2

Mean error:

$$E[error] \approx 0$$

You might incorrectly conclude:

"The model has no error."

But that's wrong.

Positive and negative errors cancel.

Variance/MSE reveals the spread.

Mean error

↓

Bias / systematic offset

Squared error

↓

Magnitude of mistakes

Variance

↓

Spread

This is why the distinction between **mean and variance** matters in ML.

---

34. Implementation From Scratch

For interviews, know how to implement the fundamental statistics without relying on NumPy.

```
def mean(values):
    return sum(values) / len(values)



def variance(values):
    mu = mean(values)
    return sum((x - mu) ** 2 for x in values) / len(values)



def std(values):
    return variance(values) ** 0.5



def covariance(x, y):
    mean_x = mean(x)
    mean_y = mean(y)

    return sum(
```

```
        (a - mean_x) * (b - mean_y)
        for a, b in zip(x, y)
    ) / len(x)


def correlation(x, y):
    cov = covariance(x, y)
    return cov / (std(x) * std(y))
```

Example

```
x = [1, 2, 3, 4, 5]
y = [2, 4, 6, 8, 10]

print(mean(x))
print(variance(x))
print(std(x))
print(covariance(x, y))
print(correlation(x, y))
```

You should understand **what every line is doing**, not merely memorize the code.

---

35. LLN Implementation

A simple simulation:

```
import random

n = 10000
total = 0

for i in range(1, n + 1):
    roll = random.randint(1, 6)
```

```
    total += roll

    average = total / i

    if i in [10, 100, 1000, 10000]:
        print(i, average)
```

Expected output will fluctuate, but generally move toward:

$$3.5$$

This directly demonstrates the Law of Large Numbers.

---

36. Interview Questions You Should Master

P0 Questions

Q1. What is expected value?

Answer:

Expected value is the probability-weighted average of a random variable's outcomes and represents its long-run average.

Q2. Does expected value have to be an actual possible outcome?

No.

Example:

X = 0 with probability 0.5

X = 10 with probability 0.5

$$E[X] = 5$$

But X never equals 5.

Q3. What is variance?

Variance measures the average squared deviation from the mean.

$$Var(X) = E[(X - \mu)^2]$$

Q4. Why square the deviations?

Because otherwise positive and negative deviations cancel.

Q5. What is the shortcut formula?

$$Var(X) = E[X^2] - E[X]^2$$

Q6. What is covariance?

Covariance measures how two random variables vary together.

$$Cov(X, Y) = E[XY] - E[X]E[Y]$$

Q7. Covariance vs correlation?

Covariance measures joint variation and depends on scale. Correlation normalizes covariance and is unitless, ranging from −1 to +1.

Q8. Does zero correlation imply independence?

No.

The source specifically provides a counterexample where $Y = X^2$, covariance is zero, but X and Y are not independent.

Q9. Does independence imply zero covariance?

Yes, under the relevant finite-moment conditions.

$$X \perp Y \Rightarrow Cov(X, Y) = 0$$

Q10. What is LLN?

As the number of independent observations grows, their sample average approaches the population expected value under the assumptions of the law.

Q11. Weak vs strong LLN?

Weak LLN describes convergence in probability; strong LLN describes almost-sure convergence.

Q12. What is MLE?

MLE chooses the parameter value that maximizes the likelihood of the observed data.

$$\hat{\theta} = \arg\max_{\theta} L(\theta)$$

or equivalently:

$$\hat{\theta} = \arg\max_{\theta} \log L\,(\theta)$$

Q13. Why use log-likelihood?

Three major reasons:

1. Converts products to sums.
2. Makes derivatives easier.
3. Improves numerical stability.

Q14. Why does maximizing likelihood equal maximizing log-likelihood?

Because log is strictly increasing.

$$\arg\max L\,(\theta) = \arg\max \log L\,(\theta)$$

---

37. The 10 Formulas to Memorize

If your interview is close, memorize these first:

1. Expected value

$$\boxed{E[X] = \sum_{x} x\,P(X = x)}$$

2. Linearity

$$\boxed{E[aX + bY] = aE[X] + bE[Y]}$$

3. Independent product

$$\boxed{E[XY] = E[X]E[Y]}$$

4. Variance

$$\boxed{Var(X) = E[(X - \mu)^2]}$$

5. Variance shortcut

$$\boxed{Var(X) = E[X^2] - E[X]^2}$$

6. Standard deviation

$$\boxed{\sigma = \sqrt{Var(X)}}$$

7. Covariance

$$\boxed{Cov(X, Y) = E[XY] - E[X]E[Y]}$$

8. Correlation

$$\boxed{\rho = \frac{Cov(X, Y)}{\sigma_X \sigma_Y}}$$

9. Chebyshev

$$\boxed{P(|\ X - \mu\ | \geq t) \leq \frac{\sigma^2}{t^2}}$$

10. MLE

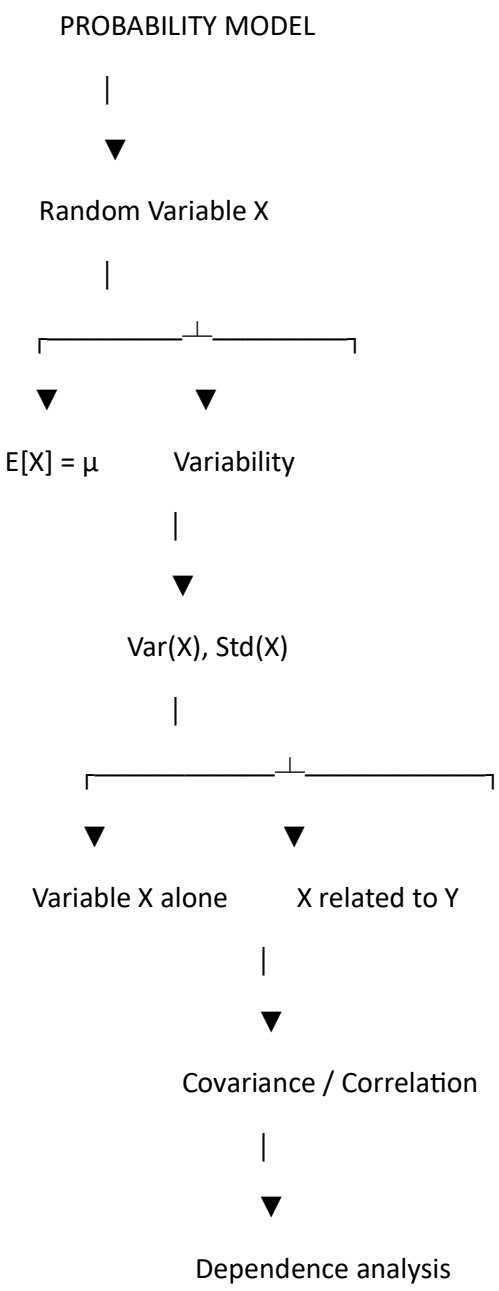
$$\boxed{\hat{\theta} = \arg\max_{\theta} L(\theta)}$$

---

38. Final Mental Model

Don't memorize this chapter as 20 disconnected definitions.

Understand it as one story:



Repeated observations



Sample average

|

▼

Law of Large Numbers

|

▼

approaches μ

Probability distribution

|

▼

Observed data

|

▼

Likelihood

|

▼

Log-likelihood

|

▼

Optimization

|

▼

MLE

|

▼

Estimated parameters

|

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ML model

If you have very limited interview time

Study in exactly this order:

Day/Session 1 — P0

1. Expected value
2. Linearity of expectation
3. Variance
4. Standard deviation
5. Covariance
6. Correlation
7. Independence vs zero correlation
8. Law of Large Numbers

Day/Session 2 — ML connection

9. Entropy
10. MLE
11. Likelihood vs probability
12. Log-likelihood
13. MSE as expected squared error
14. Implement mean/variance/covariance/correlation
15. Implement coin-toss MLE
16. Simulate LLN

That gives you the **highest-value 20% of this material** for an ML-oriented interview while still giving you enough mathematical understanding to explain *why* the formulas work.